

Lab 1 - PECARN TBI Data, STAT 214, Spring 2026

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Introduction

Traumatic Brain Injury (TBI) is among the top three causes of death and disability in children worldwide. In emergency departments, when a child is brought in for head trauma, clinicians are forced to make the critical decision of whether or not to order a CT scan. Though it is the gold standard for detecting intracranial injuries, it also exposes children to ionizing radiation and cancer risks, posing a critical tension.

PECARN pediatric head trauma dataset is a large prospective study dataset collected by Kuppermann et al. (2009) to address this critical problem. The dataset was developed to create clinical decision rules to identify children who are very low risk of clinically important TBI (ciTBI), defined as death, neurosurgery, prolonged intubation, and prolonged hospital admission related to TBI.

This dataset is important to analyze because it represents a real-world problem in which clinicians are forced to make critical decisions and must balance the risks of over-imaging and under-diagnosing. Analyzing the underlying structure of the dataset will help in better understanding the influence of clinical features, demographic features, and modeling and cleaning judgement calls that affect the predictions of clinically important TBI and recommendations for CT scans.

In my report, I will perform a structured exploratory data analysis of the PECARN dataset, and then will implement and evaluate three different classification methods: the original PECARN clinical decision rule, logistic regression, and random forest. While the objective of the analysis is to evaluate the predictive accuracy of the methods, I will also focus on the stability and interpretability of the methods, as might be required in a clinical setting.

Data

The data used in this report was collected by the PECARN study. It consists of over 43,000 cases of pediatric emergency department visits where head trauma was evaluated. Each observation in the dataset represents one patient containing features such as their demographics, injury mechanism, symptoms, Glasgow Coma Scale components, examination findings, CT scans indicators and results (when applicable), and outcomes.

The primary outcome of interest in this dataset is `PosIntFinal`, which represents clinically important traumatic brain injury (ciTBI) as defined in Kuppermann et al. The secondary outcomes of interest are `PosCT`, `DeathTBI`, `HospHead`, `Intub24Head`, and `Neurosurgery`, which represent the clinical severity spectrum of ciTBI.

Data Collection

The dataset comes from a large, prospective cohort study at 25 emergency departments in North America. Children with head injuries were enrolled and assessed using standard protocols, allowing symptom reporting to be structured. Clinicians recorded details at the bedside, such as how the injury happened, symptoms like vomiting, seizures, and dizziness, and physical exam findings. Outcomes were tracked through follow-up visits and imaging results.

The variables measured in the data set differ in terms of scale and type. Some variables are binary indicators for instance, whether a patient vomited or not, others are ordinal categories like injury severity levels, and some were continuous measures (age). Several variables were also linked in a parent-child dynamic. These differences in scale and structure require careful preprocessing before modeling.

Data Cleaning

Missingness Assessment and Indicator Creation

Missing variables were handled through an iterative process. The aim is to differentiate among three fundamentally different concepts: true absence of information, missingness due to structure, and missingness that potentially contains information.

Stage 1: Informative Missingness (True NA Only) I began by analyzing organically missing (NaN) values only. Structured codes such as 92 were intentionally left unchanged at this stage so that clinical inapplicability is not mistaken with the organic missing values during risk assessment.

For variables with missing proportion greater than 2%, I computed:

$$\text{Risk Ratio} = \frac{P(\text{ciTBI} \mid \text{Missing})}{P(\text{ciTBI} \mid \text{Observed})}$$

Variables were flagged if:

- Missing proportion > 2%, and
- Risk ratio > 2.

The following variables met these criteria:

- Dizzy
- ActNorm
- Amnesia_verb
- LOCSeparate
- Seiz
- VomitLast
- LocLen
- HASeverity

Meeting the criteria for informative missingness, I created a binary missingness indicator for each of these variable. For example, `Amnesia_verb_missing` = 1 if missing, 0 otherwise. Importantly, I did not impute the original values. They were retained as NaN.

This was an intentional modeling decision. For the data set, missing values for variables such as `Amnesia_verb` or `LOCSeparate` may actually represent clinically important states, not non-responses. Likewise, missing values for parent-level variables such as `Dizzy`, `ActNorm`, or `Seiz` may actually represent situations where a symptom could not be assessed.

Imputing these fields with the mode would have artificially introduce a negative finding when a state was not clinically defined. The missingness indicator directly captures the pattern.

In an earlier version of this pipeline, I initially imputed these variables after flagging them, under the assumption that retaining the indicator would preserve the signal. Upon reflection, I realized this was incorrect. When both the imputed value and its indicator are present, the model may partially rely on a value that does not reflect real-world clinical meaning. The current pipeline keeps only the binary indicator and removes the original values. Also, tree-based models handle missing values natively, so imputation is also unnecessary in this sense.

Stage 2: Structural Detail Variables (Code 92) Many PECARN variables are hierarchically defined. Child detail variables are only meaningful if the corresponding parent symptom is present. For example:

- VomitNbr, VomitStart, VomitLast
- SeizOccur, SeizLen
- HASeverity, HASStart
- AMSAgitated, AMSSleep, AMSSlow, AMSRepeat, AMSOth
- Palpation and clavicle detail variables (e.g., SFxPalpDepress, ClavFace, ClavNeck)

In these fields, code 92 indicates “Not applicable,” not a missing measurement. Before any transformation, all parent–child logical relationships were validated to confirm that structural consistency was preserved.

Only after this validation step were all instances of 92 converted to `NaN`. Structural detail variables were then excluded from further imputation. Their missingness reflects clinical logic (parent absent $\Rightarrow$ child undefined), not incomplete data capture. Imputing these values would collapse the hierarchical structure and assign artificial clinical detail.

Outcome Handling The primary outcome (`PosIntFinal`) was never imputed under any circumstance. Rows with missing primary outcome values were dropped before moving forward in cleaning.

Secondary outcomes (`DeathTBI`, `HospHead`, `Intub24Head`, `Neurosurgery`) were evaluated for missingness. Because total missingness across these variables remained below 1%, affected rows were dropped rather than imputed. Also, imputing severe clinical outcomes would not be defensible.

Summary No clinical predictor was imputed in the final dataset. The cleaning pipeline consisted of three iterative steps:

1. Addition of binary missingness indicators for variables with informative missingness (missing proportion $> 2\%$, risk ratio > 2), while retaining original `NaN` values.
2. Conversion of structural detail code 92 to `NaN` after logical validation.
3. Removal of observations with missing outcome values.

This approach maintains the hierarchy inherent in the PECARN data collection methodology, avoids overfitting due to imputation, and ensures that the model fits the actual data rather than an imputed version. It also ensured the data resembled the real-world as remained as much as possible.

Data Exploration

Before making judgment calls regarding data cleaning, I first did an exploratory review of the raw data to understand its clinical and statistical structure. The goal at this stage was not inference, but simply orientation: what is this data really like, and what problems does it pose?

Outcome Imbalance and Clinical Severity

An early summary of `PosIntFinal` revealed extreme class imbalance. Approximately 98% of children in the cohort did not experience clinically important traumatic brain injury (ciTBI), while only about 1.7% did.

This already changes the perspective on the problem, preventing using modeling accuracy for evaluation as it would be completely misleading. A model that simply predicted “no TBI” for all examples would already be over 98% accurate. However, the ciTBI is a very serious outcome. The fact that it is a rare outcome does not decrease significance. This tension between statistical imbalance and clinical consequence defines the predictive challenge posed in the PECARN dataset.

CT Utilization Relative to Severe Outcomes

Initial summaries also revealed a large gap between CT utilization and severe outcome. A considerable number of children received CT scans, yet only a small fraction were ultimately diagnosed with ciTBI.

Among children who received CT scans, the majority had negative results. This pattern immediately highlights the motivation underlying the PECARN study: identifying children at very low risk of ciTBI in order to safely reduce unnecessary radiation exposure.

Even before cleaning, the raw proportions suggest that imaging decisions are conservative relative to the true incidence of severe injury outcome.

Symptom-Stratified Imaging Patterns

To better understand physician decision-making, I examined CT utilization conditional on presenting symptoms:

$$P(\text{CT} = 1 \mid \text{Symptom} = 1)$$

Several ordered gradients were visible in the raw data:

- Very high CT rates for severe neurological compromise such as intubation, seizure, and altered mental status.
- Moderately high CT rates for common symptoms such as vomiting, headache, and dizziness.
- Lower CT rates when children were described as acting normally.

Even without formal modeling, these patterns suggest that imaging decisions are structured and clinically responsive rather than arbitrary.

Age Structure and Developmental Considerations

Because PECARN derives separate prediction rules for children under two years of age, age stratification is clinically meaningful. Initial exploration confirmed that the dataset encompasses the entire pediatric spectrum (no age-related outliers) and that a large number of pediatric cases include infants and toddlers.

This also provided early insight into missingness patterns. Certain symptom variables (for example, verbal amnesia) confirmed developmental restrictions, revealing that absence of data may reflect age-related inability to report rather than data entry error. This observation later informed the structured handling of missingness during cleaning.

Exploratory Takeaways

The initial exploration clarified several structural characteristics of the raw dataset:

- ciTBI is rare but clinically severe.
- CT imaging is common relative to confirmed severe injury.
- CT use increases sharply in the presence of severe neurological symptoms.
- Age and developmental status meaningfully shape symptom reporting.

These findings provided background information for further validation and cleaning issues, such as class imbalance, structured missing data, and hierarchical symptom variables.

Findings

First Finding: CT Utilization Versus Diagnostic Yield by Symptom and Age Group

Figure 1 compares two quantities for severe symptom: the CT ordering rate among patients who presented with that symptom (blue) and the diagnostic yield among those who were imaged (red), where yield is defined as the percent of CT-imaged patients with TBI on CT. The grey connector line visualizes the gap between ordering and yield. A wider gap means the symptom is strongly associated with getting a CT, but the scan rarely confirms TBI.

Results are split by age group to match the PECARN clinical decision rule structure. For children under 2 years, verbal symptoms were not assessed, so the panel focuses on neurologic and physical exam indicators. For children 2 years and older, the full symptom set is shown, including verbal symptoms and severe headache.

Two patterns stand out. First, the highest-yield findings are also some of the most targeted. In both age groups, basilar skull fracture signs and palpable skull fracture show the highest yields, and they also have relatively high CT ordering rates. This is the case where CT use appears most efficient, since a large fraction of scans are positive when these signs are present.

Second, several symptoms trigger high CT ordering despite very low yield. In children 2 years and older, neurological deficit and amnesia have some of the widest gaps, with CT ordering rates that are high but yields that remain close to the baseline. Severe headache also shows a large ordering to yield gap compared to headache (any), which is more discriminative by design in the PECARN rule. In children under 2, symptoms like not acting normally and loss of consciousness show high CT ordering but relatively low yield compared to fracture signs.

Overall, this figure highlights where CT use appears well-aligned with diagnostic confirmation and where it does not. The symptoms with large ordering to yield gaps are especially important for modeling because they represent features that may drive clinical imaging decisions without proportionally identifying TBI on CT.

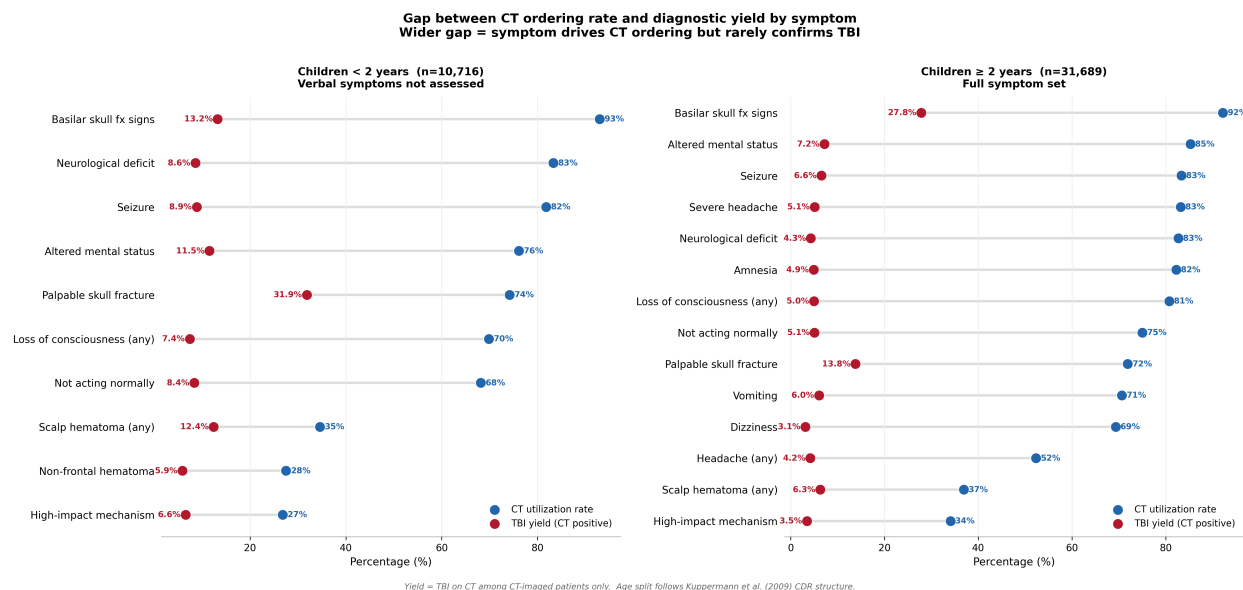


Figure 1: CT ordering rate (blue) and diagnostic yield among CT-imaged patients (red) by symptom, stratified by age group following Kuppermann et al. (2009). Grey lines connect ordering rate to yield.

Second Finding: CT Utilization Differs Substantially by Race While Yield Differences Are Smaller

Figure 2 compares CT utilization and CT-confirmed TBI yield across three racial groups: White, Black, and Other. Race was originally coded as six categories in the PECARN dataset: White (1), Black (2), Asian (3), American Indian or Alaskan Native (4), Pacific Islander (5), and Other (90). Because categories 3, 4, and 5 contained relatively small sample sizes, these groups were combined with the original “Other” category to form a single aggregated “Other” group for analysis. Patients with missing race information (n = 3,204) were excluded from this analysis.

Each bar represents 100 children within a group and is partitioned into those who received no CT, those who received a CT with a negative result, and those who received a CT with a positive TBI finding.

CT utilization varies markedly across groups. White children were imaged 42.6% of the time, compared to 35.3% for Other children and 28.2% for Black children. The White–Black difference in CT ordering is approximately 14 percentage points.

Diagnostic yield among those imaged also differs across groups, though the magnitude of these differences is smaller. Yield was 7.6% for White children, 8.4% for Other children, and 5.8% for Black children, with 95% Wilson confidence intervals shown in the figure. A chi-square test comparing yield across groups produced $p < 0.001$.

Although statistically significant, the absolute differences in yield are on the order of 2 to 3 percentage points. In contrast, the CT ordering gap between White and Black children exceeds 14 percentage points. This indicates that variation in imaging frequency is much larger than variation in CT-confirmed TBI prevalence.

In practical terms, Black children are imaged substantially less often, yet the probability that a CT scan confirms TBI among those imaged remains within a similar range across groups. The disparity in CT utilization therefore exceeds what would be expected based solely on differences in CT-confirmed TBI yield.

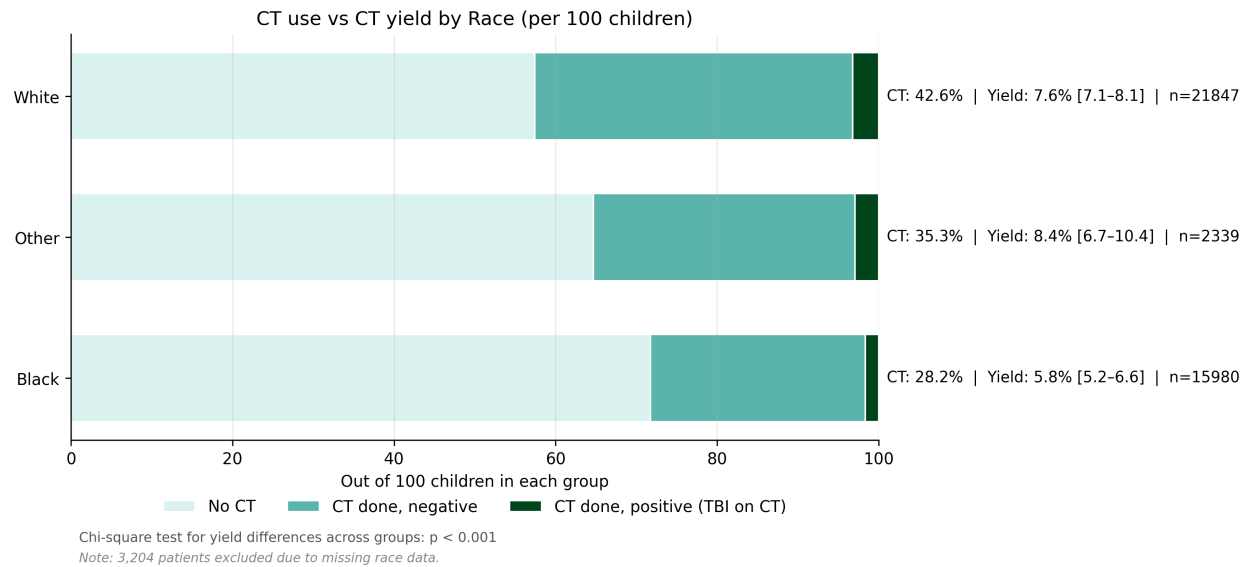


Figure 2: CT utilization and CT-confirmed TBI yield per 100 children by race. Bars are partitioned into no CT (light), CT negative (medium), and CT positive (dark). Annotations display CT ordering rate, yield with 95% Wilson confidence intervals, and group sample size. Chi-square test compares yield proportions across racial groups.

Third Finding: Racial Disparities in CT Utilization Persist After Accounting for Clinical Severity

To evaluate whether differences in CT utilization could be explained by differences in clinical presentation, I constructed a set of neurologic severity indicators based on clinical examination finds and injury characteristics. These included altered mental status, loss of consciousness, high-impact injury mechanism (`High_impact_InjSev` = 3), neurological deficit, basilar skull fracture signs, and post-traumatic seizure.

These indicators were not directly derived from the final PECARN clinical decision rule but they were present before any decisions about ordering CT scans took place. These measures are very similar to the neurologic risk factors that were used in the PECARN derivation study (Kuppermann et al., 2009), and therefore they can be considered an educated estimate of the severity of injury that is not dependent upon decisions about ordering CT scans.

Figure 3 compares White and Black children across clinical severity indicators and CT utilization rates. Differences in severity markers between groups are relatively modest, with most percentage-point gaps falling within a narrow range. For example, neurological deficit, post-traumatic seizure, and basilar skull fracture all show differences on the order of less than 1 percentage point.

Despite this similarity in clinical presentation, the use of CT scans was still significantly higher in White children. The estimated difference in CT utilization was +14.3 percentage points (White minus Black), with a 95% confidence interval excluding zero and a highly statistically significant p-value. This pattern indicates that the disparity in imaging cannot be explained solely by observed neurologic severity.

These results suggest that Black children are less likely to receive CT imaging even when they have similar clinical indicators of traumatic brain injury risk.

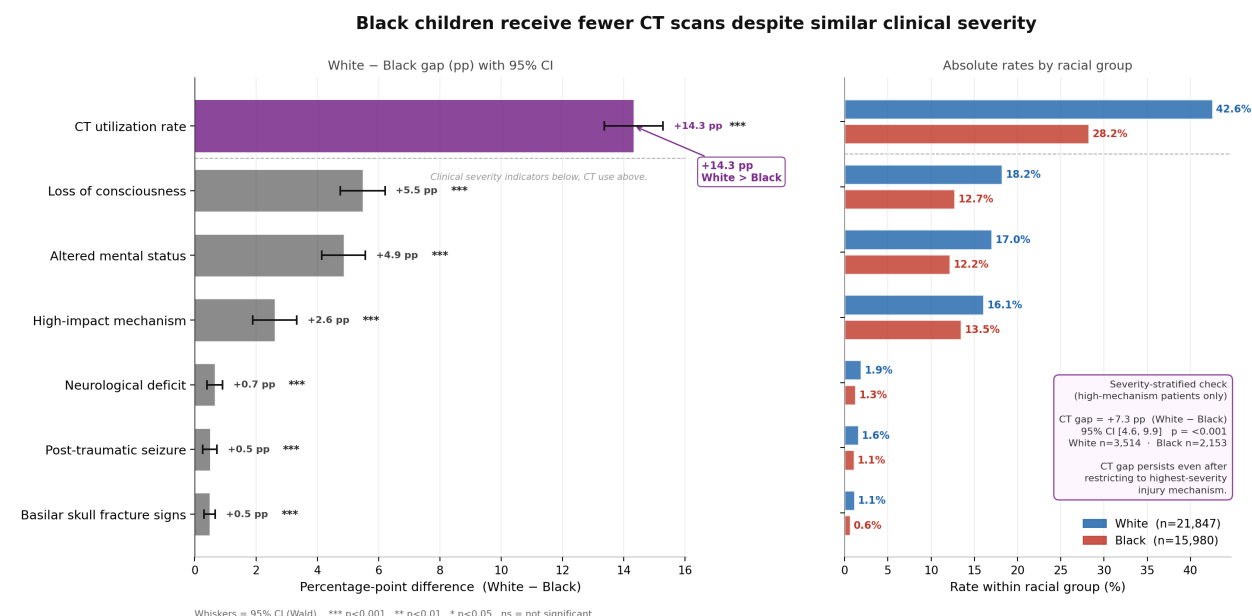


Figure 3: White-Black differences in clinical severity indicators and CT utilization. Left panel shows percentage-point differences (White minus Black) with 95% confidence intervals; positive values indicate higher rates among White children. Right panel displays absolute rates within each racial group. Differences in clinical severity markers are modest, whereas the CT utilization gap is substantially larger (approximately 14 percentage points), indicating that imaging differences exceed observed differences in measured neurologic severity.

Reality Check

A reality check applicable to the domain can be developed by comparing the patterns found in the cleaned dataset with established clinical logic and existing evidence related to decision-making in diagnosis. In pediatric TBI, the decision to perform a CT scan should be influenced by neurologic risk factors such as altered mental status, loss of consciousness, high-impact injury mechanism, seizure, and skull fracture.

In my cleaned data, the prevalence of neurologic severity indicators and their relationship with increased rates of CT usage are consistent with expectations. However, the gap in the rate of CT usage for racial groups is much larger than the corresponding gap in clinical severity indicators. Based on the assumption that the indicators used are reasonable proxies for neurologic risk and are used prior to making a decision for a CT scan, one might expect the gap in the rate of CT usage to parallel the gap in clinical severity indicators.

Existing literature on healthcare disparities supports the plausibility of such imaging disparities that are not fully explained by clinical severity. For example, research on racial and ethnic disparities in emergency care highlights that structural factors has pointed out the role of structural factors in clinical decision-making can influence differential testing patterns even when patients present with similar levels of risk (MaciasKonstantopoulos2023). The observed disparity in CT utilization by race, therefore falls within the range of patterns documented in clinical systems, suggesting that the cleaned data reflect plausible real-world behavior.

That said, this analysis does not establish causality, factors like socioeconomic status, insurance status, hospital site effects, and clinician effects may also have contributed to the observed disparities. Overall, the cleaned data passed this reality check in the sense that it is consistent with both clinical logic and existing empirical evidence regarding decision-making in diagnosis.

Stability Check

The goal of this stability check is to make sure one of my findings is not an artifact of a single judgement call in my coding choices. I focus on Finding 3 (racial disparity in CT utilization relative to clinical severity) because it relies on how I defined the “high-impact mechanism” variable.

Perturbation: redefining “high-impact mechanism”

In my main analysis, I defined “high-impact mechanism” as `High_impact_InjSev = 3` (High). As a perturbation, I relaxed the definition and treated both Moderate and High as “high-impact”:

$$\text{High_impact_InjSev} \in \{2, 3\}.$$

Everything else in the pipeline was held fixed. The point of this perturbation is that “high-impact mechanism” is an ordinal clinical severity variable, so the exact cutoff is a reasonable place where a small change could potentially change the story.

Before vs after

Under this perturbed definition Figure 4, the White–Black CT utilization gap remained substantially wider than the observed gaps in clinical severity indicators.

As expected, redefining high-impact mechanism increased the overall prevalence of this variable in both groups (White: 83.9%, Black: 82.4%), but the racial disparity in severity is still modest compared to the imaging gap. The fact that the CT gap remains constant under this more general definition of severity suggests that the result is not particularly sensitive to the choice of threshold for injury mechanism classification.

In light of this perturbation, it is clear that Finding 3 is robust. The racial disparity in CT use is not due to an arbitrary coding choice and holds under a more general specification of injury severity.

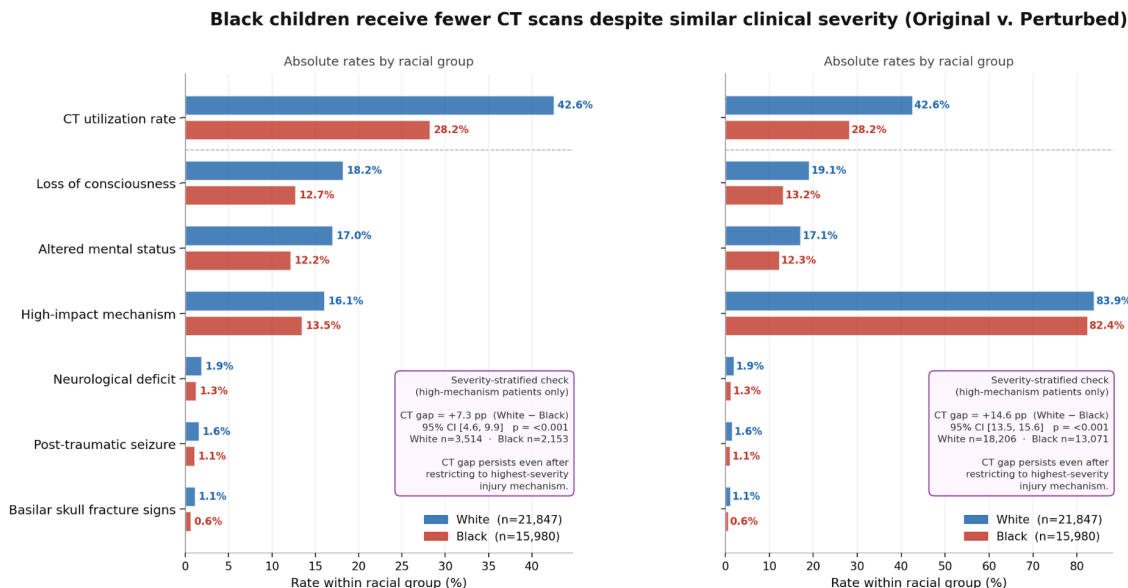


Figure 4: Stability check for Finding 3 under a perturbation of the “high-impact mechanism” definition. **Before (left):** high-impact defined as `High_impact_InjSev = 3`. **After (right):** high-impact defined as `High_impact_InjSev ∈ {2,3}`. The CT utilization gap (White–Black) persists under both specifications, indicating the finding is stable to this reasonable perturbation.

Modeling

Implementation

Three models were implemented to predict clinically-important traumatic brain injury (ciTBI): the PECARN clinical decision rule (CDR), logistic regression, and random forest.

PECARN Clinical Decision Rule (CDR). The CDR was recreated based on Kuppermann et al. (2009) through the use of the stratified approach based on the ages (separate rules for under 2 and at least 2) in Figures 2A and 2B. No training was needed. The rule was applied if at least one “high risk predictor” was present.

Logistic Regression. Logistic regression was applied based on the derived clinical features from the PECARN rule. In an alternate implementation, an EDA-derived set of clinical features was used. All features are converted to a numeric type. Missing values are replaced with zeros if they are “present,” otherwise they are ignored. A `StandardScaler` was applied before model fitting to ensure coefficient stability. L2 regularization was used (default $C = 1.0$), and `class_weight="balanced"` was specified to address class imbalance, as ciTBI occurs in approximately 1% of cases.

Random Forest. The random forest model was fit with 300 trees, maximum depth 8, and minimum leaf size 20. Class weights were balanced. Rather than using the default 0.5 threshold, the classification threshold was tuned on a validation set to achieve a target sensitivity of 0.85 (default 0.95) while maintaining at least minimal specificity. This mirrors the clinical priority of minimizing missed injuries.

Data Splitting. Data were split into 60% training, 20% validation, and 20% test sets using stratified sampling on the outcome variable to preserve ciTBI prevalence across splits.

Performance was evaluated on the held-out test set using sensitivity, specificity, positive predictive value

(PPV), negative predictive value (NPV), ROC AUC (for probabilistic models), and implied CT rate (fraction of patients flagged as high risk).

Interpretability

The PECARN CDR is fully interpretable. It consists of clear if-then rules obtained through recursive partitioning and corresponds directly to clinical observations.

Logistic regression is also interpretable. Each variable is assigned a coefficient that describes its contribution to the log-odds of ciTBI. Positive coefficients increase predicted risk, while negative coefficients decrease it. Because features were standardized prior to fitting, coefficient magnitudes are directly comparable.

The random forest is not inherently interpretable due to its ensemble structure. However, feature importances were extracted to understand which predictors most influenced classification decisions. Threshold tuning also helps to explain how the model balances sensitivity and specificity.

In all models, sensitivity is the most important clinical metric because false negatives indicate a missed traumatic brain injury.

Stability

To evaluate robustness in the spirit of PCS, I examined stability along three dimensions:

1. preprocessing (cleaning) perturbations
2. feature set specification
3. age stratification.

The stability analysis in the modeling stage is similar to the EDA section above but at a different level of uncertainty. In the EDA section above, the focus of the perturbations was focused on how clinical severity is encoded in the dataset. The question was whether conclusions about the descriptive statistics of the data would change if different severity definitions were used. However, In the modeling stage, after the encoding options have been determined, the source of uncertainty shifts to modeling specification. Stability is then assessed in terms of whether conclusions derived from the modeling stage are dependent on the feature sets and age stratification.

Cleaning stability

Two perturbations were introduced to the preprocessing pipeline:

1. **Missingness threshold perturbation:** increasing the feature-screening cutoff from 2% to 10% (RR filter on).
2. **Informative-missingness perturbation:** disabling the risk-ratio (RR) filter used to flag variables whose organically-missing values were associated with elevated ciTBI risk (threshold fixed at 2%).

The 2% cutoff represents a conservative screening rule, while 10% is significantly more lenient. Removing the RR filter corresponds to dropping a protection against potentially informative missingness. In both instances, the subsequent modeling pipeline (set of features, data splits, hyperparameter tuning, and evaluation criteria) remained unchanged.

As shown in Figure 5, sensitivity, specificity, negative predictive value (NPV), and implied CT rate were unchanged across all three models under both perturbations:

CDR: (0.960, 0.569, 0.435, 0.999) unchanged
LR: (0.733, 0.824, 0.181, 0.997) unchanged
RF: (0.987, 0.400, 0.603, 1.000) unchanged

Importantly, these preprocessing perturbations were performed after resolving the race-sensitive missingness issue discovered during EDA, so the assessment of model stability over these preprocessing choices is based on a stabilized representation layer.

This stability suggests a high degree of structural stability with regard to these preprocessing choices. Within the considered range, the cleaning choices did not significantly affect the model behavior or the underlying clinical tradeoffs.

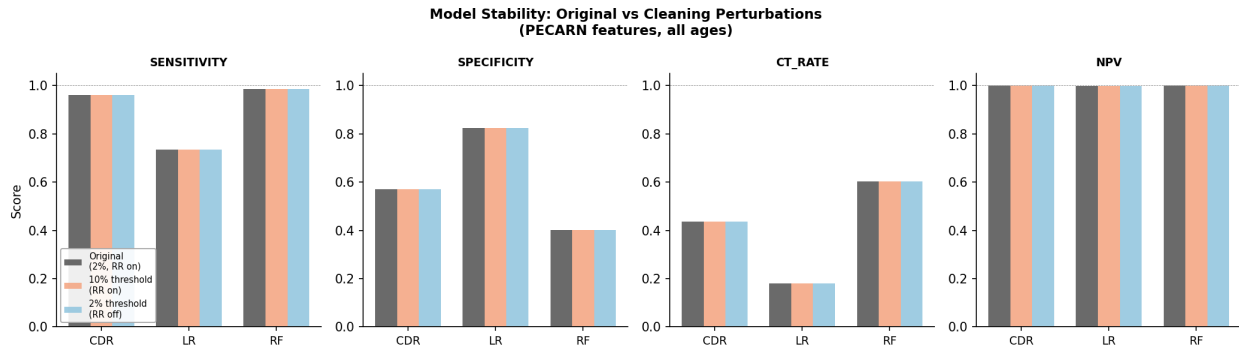


Figure 5: Model stability under cleaning perturbations (PECARN features, all ages). Performance for PECARN (CDR), logistic regression (LR), and random forest (RF) is shown under the original preprocessing rule (2% missingness threshold with RR filter on), a 10% threshold (RR on), and with the RR filter disabled (2% threshold). Sensitivity, specificity, NPV, and implied CT rate are unchanged across conditions, indicating robustness to these preprocessing choices.

Modeling specification stability

In contrast, model behavior was meaningfully sensitive to higher-level modeling decisions.

Feature set sensitivity. Switching from PECARN-derived predictors to the EDA-informed feature set altered the sensitivity–specificity tradeoff, particularly for the random forest.

For example, in the all-ages setting:

- **Random Forest (PECARN features):** sensitivity 0.987, implied CT rate 0.603.
- **Random Forest (EDA features):** sensitivity 0.800, implied CT rate 0.246.

As such, the PECARN feature set provided for extremely high sensitivity levels but with significant CT utilization levels, whereas the EDA feature set provided for a balanced profile but with lower levels of sensitivity rate. Notably, the target sensitivity constraint (0.968 for over 2, 1.00 for under 2) could not be achieved under the EDA feature set for the random forest, indicating structural differences in predictive capacity across feature representations.

For logistic regression, feature-set changes produced more modest shifts, though effects were more pronounced in the under-2 group (sensitivity 0.733 with 0.824 increased specificity).

Age stratification sensitivity. Sensitivity varied significantly across developmental groups, even when controlling for model class and feature set. Figure 6 displays sensitivity for PECARN CDR, logistic regression, and random forest across all ages combined, children under 2 years of age, and children 2+ years of age.

For PECARN CDR, sensitivity was 0.960 for all ages combined, 0.950 for children under 2 years of age, and 1.000 for children 2+ years of age. These values are similar to those used for validation targets in Kuppermann et al. (2009), which are displayed as dashed lines.

Sensitivity for the random forest was extremely high across all age groups: 0.987 for all ages combined, 0.950 for children under 2 years of age, and 1.000 for children 2+ years of age. These values all meet or exceed the 0.95 benchmark. As shown previously, however, this came at the cost of high CT use.

In contrast, sensitivity for logistic regression failed to meet or exceed 0.95 sensitivity for any age group: 0.733 for all ages combined, 0.750 for children under 2 years of age, and 0.804 for children 2+ years of age.

These results demonstrate that while the PECARN CDR and random forest can replicate or exceed benchmark sensitivity levels in specific subgroups, performance patterns still differ by developmental subgroup. Thus, model conclusions are not uniform across age groups, reinforcing that age stratification is a substantive modeling choice rather than a cosmetic one.

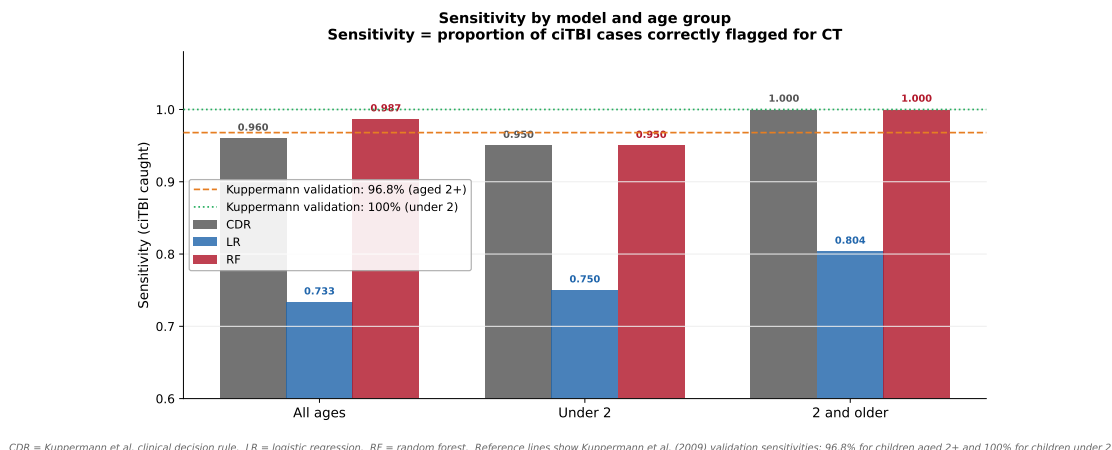


Figure 6: Sensitivity by model and age group (PECARN features). Sensitivity is shown for all ages combined, children under 2 years, and children aged 2+ years. The dashed and dotted lines indicates the target sensitivity benchmarks from Kuppermann et al. (2009).

Taken together, these findings indicate a complex pattern of stability. At the data-representation level, missingness addressed in the preprocessing perturbations did not affect predictive behavior. At the modeling-specification level, however, feature engineering and age stratification played important roles in the clinical trade-off between sensitivity and CT use. The main sources of variability in this project are not cleaning thresholds, but rather large modeling decisions about what data to include and how to stratify populations.

Discussion

The framework for this project was organized around a limited set of domain questions rather than individual statistical methods. The initial questions that I wanted to address included: How does CT usage compare to ciTBI prevalence? Next, I wanted to assess if there was alignment in terms of severity indicators used for imaging decisions. Finally, I wanted to assess if these patterns held true across racial groups and alternative modeling specifications.

Data Size and Class Imbalance

In spite of having thousands of data points in the dataset, clinically important TBI (ciTBI) occurs in only 1% of the cases. This limited signal for the primary outcome was a constraint. The sensitivity measure was the primary measure to evaluate the model because false negative rates equate to missing traumatic brain injuries. In instances of rare prevalence, small classification changes have a big impact.

The small number of positive cases also limits the extent to which complex models can learn beyond existing clinical predictors. The random forest model has very high sensitivity, but this came at the expense of increasing CT usage. This reflects a structural issue outside the scope of the algorithm.

Data and Reality

There is no one-to-one correspondence between the dataset and clinical reality. The severity is represented through discrete variables. The missing values are more related to documentation than absence. The use of CT scans is an observed phenomenon rather than ground truth.

Two types of visualizations were used. In the cleaning phase, they were used to detect inconsistencies. In the exploration phase, they were used to summarize clinically relevant information. Comparability was an important aspect. The CT scan rates are compared as proportions rather than direct counts to combat class imbalance.

Stability and Judgment Calls

Visualization and preprocessing choices are not neutral. Following the PCS framework, I checked whether the results depended on reasonable judgment calls. Changing the cleaning perturbations, such as increasing the missingness threshold from 2% to 10% and disabling the RR filter, does not affect the performance metrics of the model, suggesting structural stability in preprocessing choices.

In contrast, the higher-level modeling decisions, especially the feature set specification and the stratification by age, substantially impacted the tradeoff between sensitivity and specificity. Thus, instability arose primarily from substantive modeling decisions rather than cleaning thresholds.

Tradeoffs among Algorithms

The PECARN CDR is a rule-based clinical algorithm. Logistic regression has a linear log-odds structure, while the random forest has a nonlinear combination of decision trees. They all operate on encoded features, not clinical nuances.

This led to the explicit consideration of the tradeoff between sensitivity and specificity. High sensitivity was attainable, but with a price in terms of CT usage. The best performance in terms of sensitivity was achieved with the random forest, but with a drawback in interpretability compared with logistic regression and the PECARN CDR.

Future Data Implications

Models trained on historical decisions risk reinforcing existing patterns. If there are disparities in the use of CT among the population in the training data, then the model may replicate these disparities. Furthermore, the implementation of a highly sensitive model will change the behavior of imaging studies, and this will change the distribution of future data impacting how it represents reality.

Conclusion

This analysis has investigated pediatric TBI from descriptive, equity-focused, and predictive perspectives. These perspectives are all centered on the core domain problem that PECARN was originally designed to address: how to detect clinically important TBI without subjecting patients to unnecessary CT scanning.

The descriptive analyses highlighted a discrepancy between CT scanning rates and diagnostic yields. The rates of CT scanning were significantly higher than the prevalence of cTBI, which reflects the importance of sensitivity and avoiding harm. However, most CT scans were negative, which reflects the trade-off between safety and radiation exposure that originally led to the development of the PECARN rule.

The equity-focused analyses highlighted that there are differences in CT scanning rates between racial groups, even when adjusted for severity indicators. Most importantly, these results are stable to reasonable variations in defining severity in preprocessing. While we cannot determine the underlying reasons for such differences from this dataset alone, we can show that decision-making patterns vary in structured and measurable ways.

The modeling phase evaluated three methods: the rule-based PECARN CDR, logistic regression, and random forest. In all models, the same structural constraint was imposed: ciTBI is a rare event. The random forest method had the highest sensitivity, but failed to improve the implied CT rate. The logistic regression model produced a more conservative imaging strategy, but failed to reach benchmark levels of sensitivity. These findings simply affirm a key insight in the domain: increasing sensitivity increases imaging. Predictive algorithms do not remove this tradeoff, they formalize it.

Stability analysis revealed a layered pattern. The findings were insensitive to perturbations in preprocessing steps such as missing thresholds and informative missingness filters. In contrast, there was a more nuanced sensitivity in the predictive findings to higher-level modeling decisions, including specification of the feature set and stratification by age. Therefore, the primary drivers of analytical variability were at a higher modeling level.

In all, the PECARN dataset shows that the problem at hand cannot be solved without varying tradeoffs. The clinical decision rule is a clinically trusted algorithm not because it makes predictions under certainty, but because it puts an emphasis on not missing ciTBI as it levels it at 500-to-1 ratio of impact. The data reflect both underlying pathology and human decision-making and models trained on these data inherit that structure.

The overall takeaway is that prediction performance, image use, and equity are closely related. Any use of prediction in this space will require a balance of these factors. The PECARN framework provides a clinically grounded starting point, but quantitative modeling makes unavoidable tradeoffs inherent in pediatric emergency care.

Academic Honesty Statement

I affirm that this submission represents my own independent work. I participated in general conceptual discussions with my peers, but all analysis, coding, modeling decisions, and written interpretation were completed by me. I also used generative AI tools to help clarify conceptual gaps and assist with coding robustness.

Collaborators

This project was completed independently.

Bibliography

Macias-Konstantopoulos, W. L., Collins, K. A., Diaz, R., Duber, H. C., Edwards, C. D., Hsu, A. P., Ranney, M. L., Riviello, R. J., Wettstein, Z. S., & Sachs, C. J. (2023). Race, healthcare, and health disparities: A critical review and recommendations for advancing health equity. *Journal of the American College of Emergency Physicians Open*, 4(5), e13075. <https://doi.org/10.1002/emp2.13075>

Madley-Dowd, P., Hughes, R., Tilling, K., & Heron, J. (2019). The proportion of missing data should not be used to guide decisions on multiple imputation. *Journal of Clinical Epidemiology*, 110, 63–73. <https://doi.org/10.1016/j.jclinepi.2019.02.016>

Yu, B., & Barter, R. L. (2024). *Veridical data science: The practice of responsible data analysis and decision making* (Open access ed.). The MIT Press.